

Deep Learning

Lab 1

Tensorflow

- Biblioteca – Google
- <https://playground.tensorflow.org/>

<https://docs.anaconda.com/free/anaconda/applications/tensorflow/>

```
conda create -n tf tensorflow
```

```
conda activate tf
```



Epoch
000,083

Learning rate
0.03

Activation
Sigmoid

Regularization
None

Regularization rate
0

Problem type
Classification

DATA

Which dataset do you want to use?



Ratio of training to test data: 50%



Noise: 0



Batch size: 10



FEATURES

Which properties do you want to feed in?



+ - 1 HIDDEN LAYER

+ -

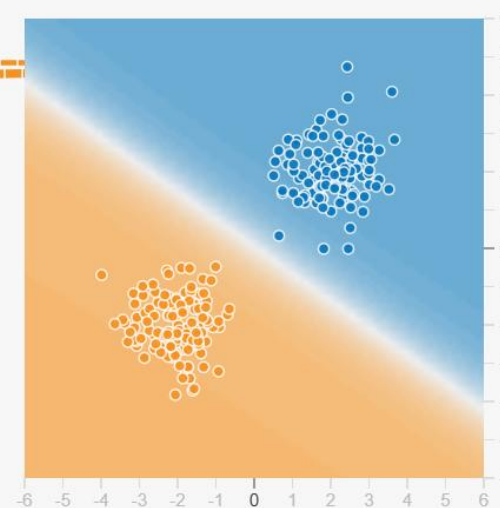
2 neurons



This is the output from one **neuron**.
Hover to see it larger.

OUTPUT

Test loss 0.005
Training loss 0.005





Epoch
000,908

Learning rate
0.03

Activation
Sigmoid

Regularization
None

Regularization rate
0

Problem type
Classification

DATA

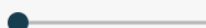
Which dataset do you want to use?



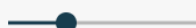
Ratio of training to test data: 50%



Noise: 0

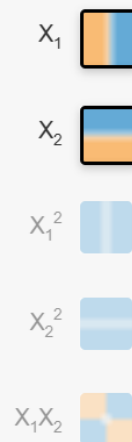


Batch size: 10



FEATURES

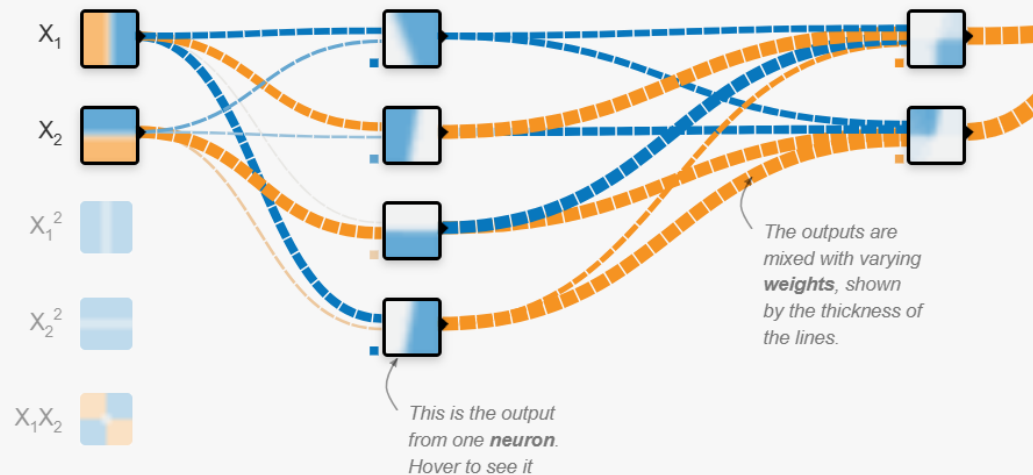
Which properties do you want to feed in?



+ - 2 HIDDEN LAYERS

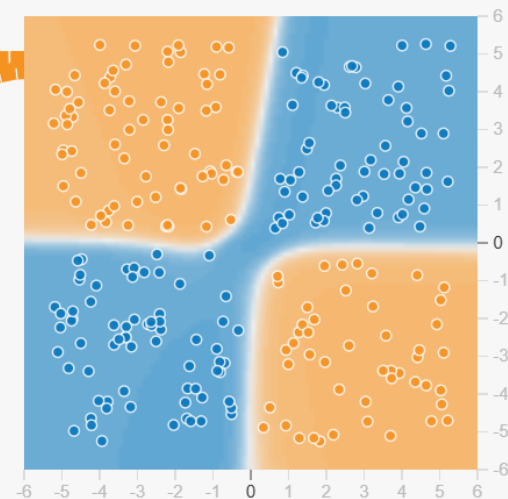
+ -
4 neurons

+ -
2 neurons



OUTPUT

Test loss 0.010
Training loss 0.005





Epoch
000,562

Learning rate
0.03

Activation
Sigmoid

Regularization
None

Regularization rate
0

Problem type
Classification

DATA

Which dataset do you want to use?



Ratio of training to test data: 50%



Noise: 0

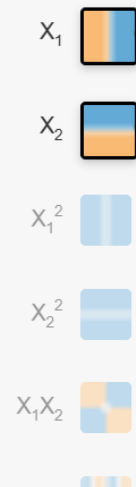


Batch size: 10



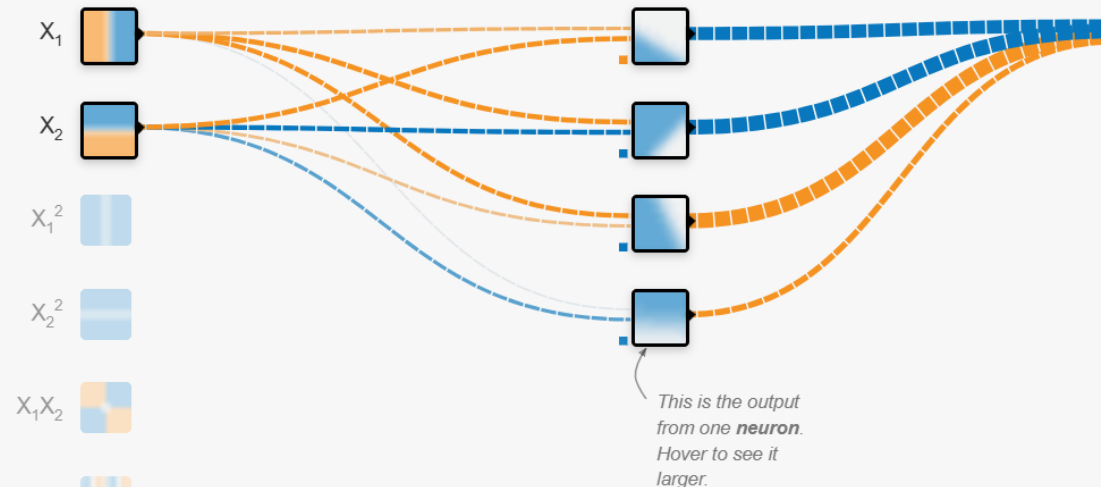
FEATURES

Which properties do you want to feed in?



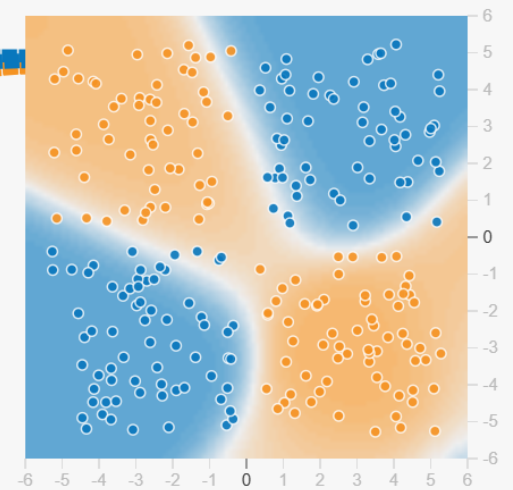
+ - 1 HIDDEN LAYER

+ -
4 neurons



OUTPUT

Test loss 0.065
Training loss 0.070



Tensor flow & keras

TensorFlow este o platformă open-source pentru machine learning și o bibliotecă matematică simbolică care este utilizată pentru aplicații de învățare automată.

Este conceput pentru a fi low-level pentru programarea diferențiabilă, care se ocupă de tensori, variabile și gradienti .

Keras este o bibliotecă open-source destinate rețelelor neuronale care rulează pe TensorFlow. Este conceput pentru a fi un API la nivel înalt pentru deep learning.

Keras oferă o interfață mai simplă și mai ușor de utilizat pentru construirea și antrenarea rețelelor neuronale, în timp ce TensorFlow vă oferă mai mult control asupra arhitecturii și procesului de instruire.

Keras servește ca API la nivel înalt pentru TensorFlow, făcând TensorFlow simplu și productiv.

Exemplu

- identificarea cifrelor scrise de mana folosind retele neuronale
- Colectia de date: MNIST

- https://www.tensorflow.org/api_docs/python/tf/keras/datasets
Module: tf.keras.datasets

Modules

`boston_housing` module

`cifar10` module

`cifar100` module

`fashion_mnist` module

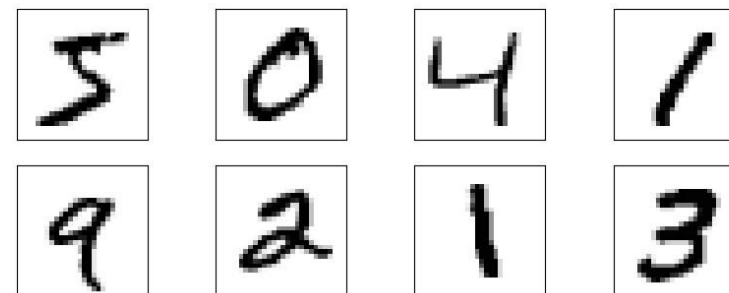
`imdb` module

`mnist` module

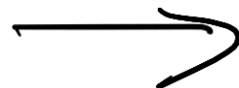
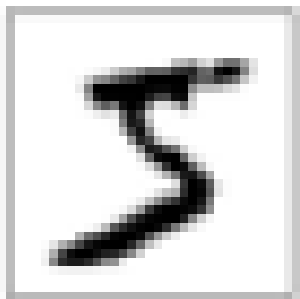
`reuters` module

MNIST

- Colectia MNIST este o colectie de date mare de imagini cu cifre scrise de mana, care este utilizata in mod obisnuit pentru antrenarea diferitelor sisteme de procesare a imaginilor.
- Baza de date contine 60.000 de imagini in mt de training si 10.000 de imagini de testare
- Aceasta este utilizata in mod obisnuit pentru antrenarea si testarea modelelor de ML pentru recunoasterea cifrelor scrise de mana.
- Aceasta este o colectie de date standard in domeniul ML.

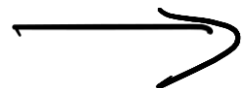
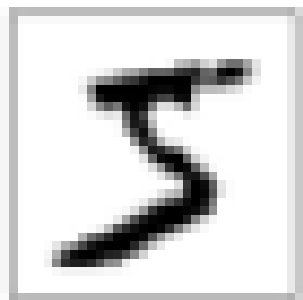


Datele de intrare sunt imagini.



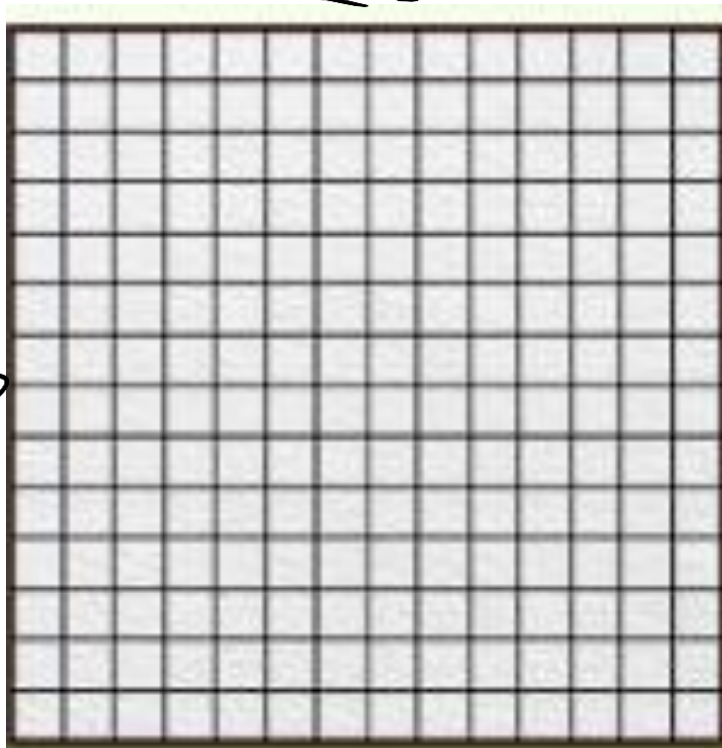
Retelele neuronale au date de intrare valori numerice.

Datele de intrare sunt imagini.

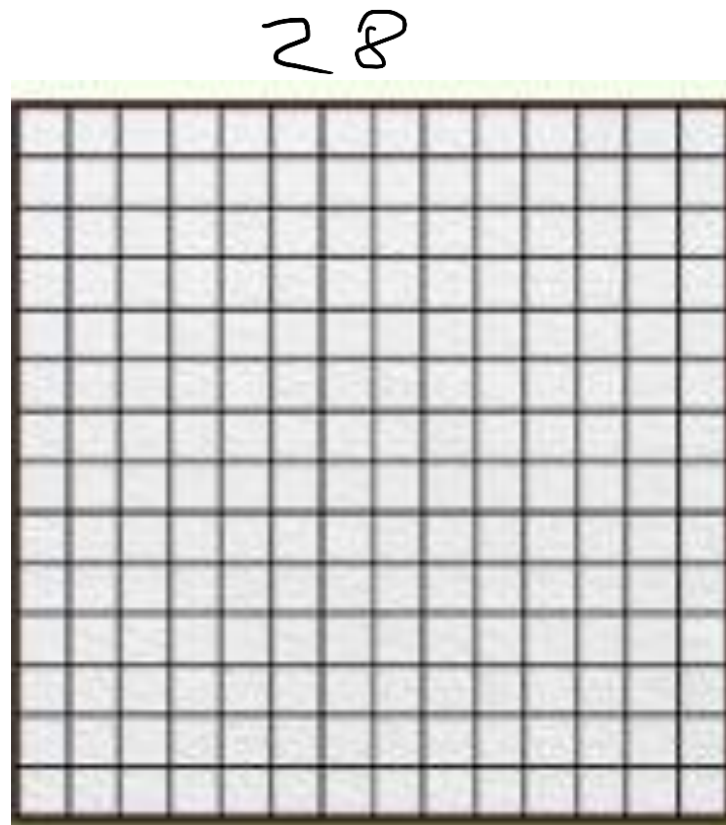
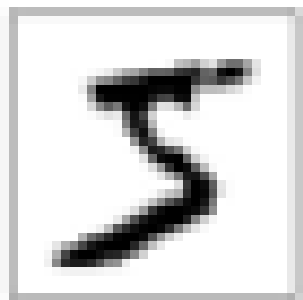


28

28



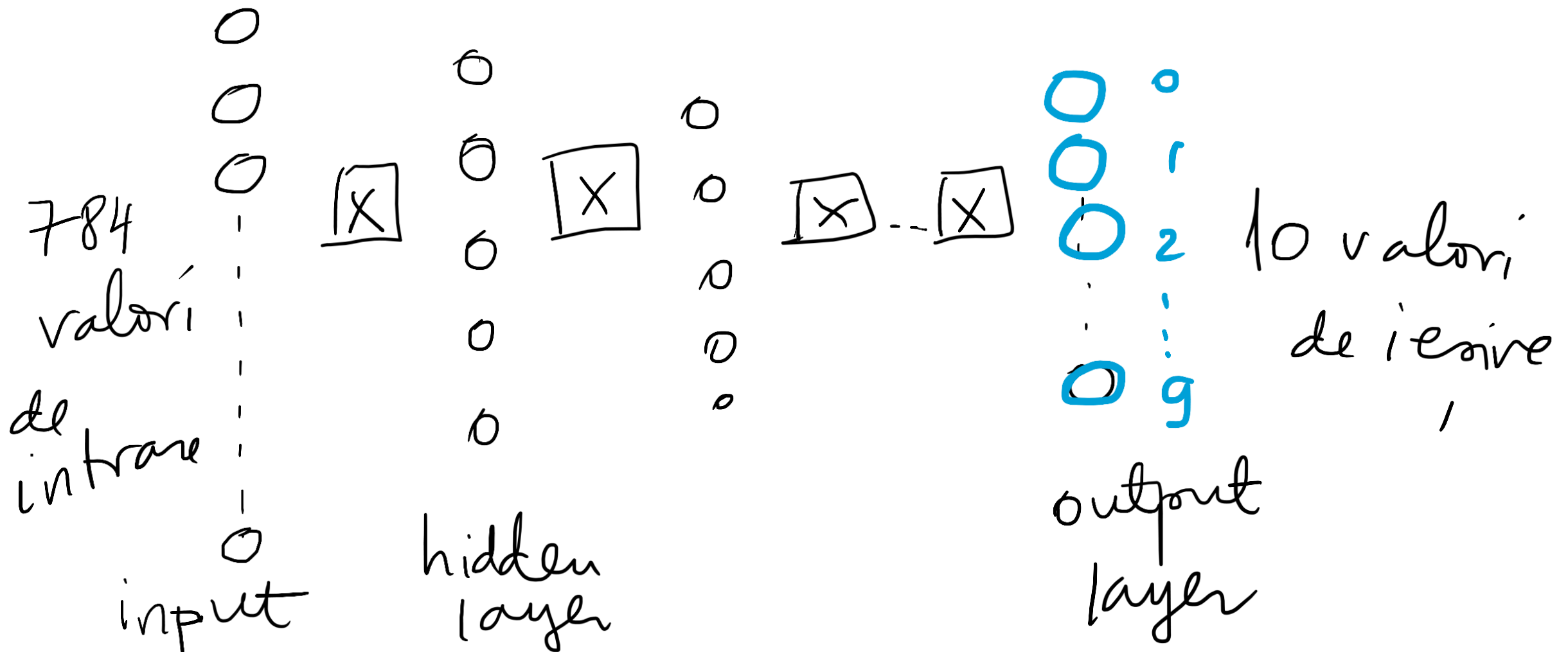
Retelele neuronale au date de intrare valori numerice.



28x28 pixeli, fiecare pixel avand o valoare $[0, 255]$ in functie de intensitatea colorii alb/negru (cat de mult al contine 255 = negru si 0 = complet alb. Altfel nuate de gri)

Fiecare valoare de intrare va fi

o imagine, $28 \times 28 = 784$ valori



Un vector de logit sau scoruri de log-odds este un vector de valori numerice care reprezinta probabilitatea logaritmica a unei anumite clase.

In loc sa prezicem direct probabilitatea unei clase, modelele de regresie logistica prezic logaritmul sansei de a apartine acelei clase.

Aceasta transformare logaritmica poate fi utila pentru a evita problemele de sub- sau supra-flux care pot aparea atunci cand se lucreaza cu probabilitati mici sau mari.

Pentru a converti un vector de logit in probabilitate, putem aplica functia softmax, care transforma valorile logit in intervalul (0, 1).

Functia softmax

$$\sigma: \mathbb{R}^k \rightarrow (0, 1)^k$$

$$(z_1, \dots, z_k)$$

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

$$\text{Obs.} \sum_{i=1}^k \sigma(z)_i = 1.$$